IR Matching of the Cosmological Constant in a Classical Vacuum-Response EFT

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The cosmological constant problem includes a 120-order mismatch between misapplied Planck-scale zero-point estimates and the observed infrared density of order $10^{-27}\,\mathrm{kg/m^3}$. This paper records a classical scalar–tensor effective field theory for a vacuum-response field and an explicit constitutive infrared matching condition linking the galactic acceleration scale a_0 to the Hubble rate H_0. We do not claim that this matching is derived by varying the action at the Hubble radius; it embeds the known modified-Newtonian-dynamics–cosmology coincidence in covariant language. We also do not derive the dark-energy fraction from a_0 alone; that quantity remains fixed by cosmic microwave background data. There is no quantum field-theory loop calculation; vacuum refers to the classical response field only. Numerical tables report internal consistency benchmarks against published Solar-System, galactic-rotation, and gravitational-wave constraints, not independent discoveries.

I. INTRODUCTION

The cosmological constant problem remains arguably the most profound crisis in modern theoretical physics [1–3]. Within the standard framework of Quantum Field Theory (QFT), the vacuum energy density is computed as the sum of zero-point energies across all fields up to a fundamental cutoff scale. If the Planck mass (M_{Pl}) is adopted as the natural ultraviolet (UV) limit, the theoretical vacuum energy density scales as $\rho_{\mathrm{th}} \sim M_{\mathrm{Pl}}^4 \sim 10^{92}$ kg/m^3. Conversely, cosmological measurements from the cosmic microwave background (CMB) and distant Type Ia Supernovae place the observed dark energy density at $\rho_{\mathrm{obs}} \sim 10^{-26}$ kg/m^3 [4–6]. This yields an unprecedented mismatch of approximately 120 orders of magnitude, frequently referred to as the cosmological constant catastrophe [8].

To reconcile this immense fine-tuning dilemma, standard ΛCDM cosmology treats Λ either as an ad-hoc geometric integration constant or introduces dynamical scalar fields (e.g., quintessence) that require delicate, unnatural potentials [9–12]. Alternative approaches attempt to exploit holographic bounds or modified gravity schemes [13–21], yet a direct analytical bridge linking microscopic vacuum parameters to macroscopic cosmic parameters without fine-tuning has remained elusive.

In this work we separate three layers that prior drafts conflated:

1. a covariant *classical* EFT for P_{vac} (Eq. (14));
2. a *constitutive* IR matching $\Omega_{\mathrm{bd}} = a_0/(cH_0)$ (Sec. IV G), numerically the same content as the long-standing MOND–H_0 coincidence [22, 23];
3. a passive ΛCDM background Ω_Λ fixed by CMB input, not output from a_0.

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The title phrase "vacuum" refers to the classical response field P_{vac}, not to a renormalized QFT vacuum energy. Inspired by the thermodynamic reading of horizons [24], we treat $R_H = c/H_0$ as an IR matching scale, not a material shell (Sec. IV A).

To address the galactic rotation curve anomaly without invoking ad-hoc, undetected dark matter particles, we propose that the physical vacuum is not a passive geometric background. Instead, it behaves as an active structured medium endowed with non-local memory and a plastic-like rheological response. Under the action of compact baryonic sources and localized core vorticity, the vacuum field undergoes a dynamical stress redistribution. This mechanism generates an extended effective mass profile that naturally accounts for asymptotically flat galactic rotation curves.

A. Observables, benchmarks, and falsifiable targets

To meet standard *Physical Review D* expectations on testability, we state explicitly which published data anchor the parameters and which outputs are falsifiable predictions versus consistency checks.

To establish the empirical validity of our classical EFT framework, we define three independent observational benchmarks. First, the spatial variation of the response field P_{vac} modifies the local geodesic equation, introducing a non-local acceleration correction. This modification must systematically reproduce the flat rotation curves of the SPARC database without inflating the stellar mass-to-light ratios.

The primary numerical benchmark is the localized memory length scale, empirically constrained to $r_m \approx 6.6$ kpc for Milky Way-like systems. This scale serves as a fixed parameter that breaks the typical halo-mass degeneracy inherent to standard dark matter parametrizations like the Navarro-Frenk-White (NFW) profile.

Furthermore, the model provides distinct, falsifiable targets at astrophysical and cosmological scales:

- **Edge velocity scale invariance:** At extreme radii, the induced velocity asymptotically approaches a common scale $v_{\rm mem,edge} \sim 100 \pm 20$ km/s, decoupled from the total baryonic mass.
- **Lensing-rotation consistency:** The effective surface density $\Sigma(R)$ derived from stress redistribution must predict gravitational lensing deflection profiles that match kinematic data without secondary modifications.
- **Environmental modulation:** As a non-local rheological response, the value of r_m is predicted to vary weakly under the influence of external tidal fields or group-scale membership.

a. Fixed observational anchors (inputs). Background cosmology uses Planck 2018 values $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.688$, and $\Omega_m = 0.315$ [4]. Recent DESI baryon-acoustic-isochronous (BAO) constraints and the classic SDSS acoustic peak [25, 26] are compatible with this passive $w = -1$ anchor at percent-level precision in $(\Omega_m h^2, H_0)$; they are cited as external checks, not refit inputs to the Planck anchors listed here. The galactic scale is $a_0 = 1.20 \times 10^{-10}\,\mathrm{m/s^2}$ (SPARC/MOND phenomenology) [22, 23, 27?]. Local gravity uses post-Newtonian parameters $\gamma = \beta = 1$ at the field minimum, consistent with Cassini $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [30, 31]. Tensor propagation uses $v_{\rm gw} = c$, compatible with GW170817–GRB 170817A [32].

b. Internal consistency benchmarks (not new discoveries). Table II and Sec. VI A compare the EFT to Mercury perihelion, SPARC-type rotation speeds, Sirius B redshift, core saturation, and binary GW speed at the level of order-of-magnitude agreement with standard benchmarks.

c. Falsifiable targets specific to this framework. (i) Horizon attractor fraction $x_\star = \Omega_m/(2\alpha_s)$ from H_0, P_0, and Gibbons–Hawking entropy *without* Ω_Λ in the χ sector (Sec. IV I; current post-diction $x_\star \simeq 0.689$ vs. Planck 0.688). (ii) Failure of volumetric three-dimensional hysteresis as a Λ estimator (null test; Sec. B). (iii) Sensitivity of $x_\star$ to the cubic entropy slope ζ/χ (Table IV). (iv) Galactic rotation from the integrated AQUAL limit of the scalar gradient law (script `test2_sparc_field.py`). A failure of any of (i)–(iii) at the quoted percent level would refute the constitutive closure as implemented; (iv) is an open numerical channel.

II. DYNAMICAL VACUUM AND NON-LOCAL MEMORY TRANSPORT

A. Conservation and Relaxation Dynamics

We model the physical vacuum as a coherent emergent substrate governed by a scalar density field $\rho(\mathbf{x}, t)$. The temporal evolution of the pressure and density states within this medium is dictated by a modified continuity equation incorporating diffusion and linear local relaxation:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) + D\nabla^2 \rho - \gamma(\rho - \rho_0) \tag{1}$$

where ρ_0 represents the background vacuum density at fundamental equilibrium, D is the effective diffusion coefficient, and γ denotes the relaxation rate. The term $-\gamma(\rho - \rho_0)$ provides the physical restoring force of the vacuum as it attempts to recover its equilibrium state after being compressed by baryonic structures. The velocity field $\mathbf{v}$ driving the phase transport is given by:

$$\mathbf{v} = \mathbf{v}_{\rm source} + \mathbf{v}_{\rm vortex} - \lambda \nabla \rho \tag{2}$$

where $\mathbf{v}_{\rm source}$ accounts for external driving forces, $\mathbf{v}_{\rm vortex}$ couples the flow to intrinsic angular momentum (core vorticity and spin), and $-\lambda\nabla\rho$ is the non-linear response to localized vacuum pressure gradients.

B. Rheological Memory Kernel and Gravitational Modification

To capture the persistent plastic-like behavior of the vacuum under gravitational deformation, the accumulated stress $\sigma(r)$ is not dissipated locally. Instead, it undergoes a non-local redistribution mediated by a spatial memory operator:

$$\mathcal{M}[\sigma](r) = \int_0^r K(r, r')\sigma(r')\, dr' \tag{3}$$

where the finite-range exponential kernel is parameterized as:

$$K(r, r') \sim \exp\left(-\frac{|r - r'|}{r_m}\right) \tag{4}$$

Here, r_m defines the characteristic spatial memory scale of the effective substrate. This rheological modification transforms the acceleration profile, mapping the non-local phase redistribution into a stable gravitational velocity support:

$$v_{\rm obs}^2(r) = v_{\rm bar}^2(r) + v_{\rm mem}^2(r) \tag{5}$$

where $v_{\rm bar}(r)$ is the standard Newtonian baryonic velocity, and $v_{\rm mem}^2(r) = M_{\rm eff}(r)/r$ provides the continuous structural support required to sustain flat profiles without unseen matter.

C. Vacuum Energy-Momentum Tensor

We consider an effective stress-energy tensor for the vacuum given by:

$$T_{\mu\nu}^{\rm vac} = -\rho_{\rm vac} c^2 g_{\mu\nu}, \tag{6}$$

where $\rho_{\rm vac}$ represents the coherent vacuum energy density. The Einstein field equations assume the standard form:

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu}^{\rm matter} + T_{\mu\nu}^{\rm vac}\right). \tag{7}$$

By defining the effective cosmological constant as $\Lambda = \frac{8\pi G}{c^2}\rho_{\rm vac}$, the unified field equation reduces naturally to:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\rm matter}. \tag{8}$$

D. Lorentz Invariance and Energy-Momentum Conservation

The vacuum sector is encoded by a *Lorentz scalar* energy density ρ_Λ at the passive minimum $P_{\rm vac} = P_0$ (fixed by the cromatic2/Planck ΛCDM background), while the dynamical boundary piece $\rho_{\rm bd}$ of Eq. (30) is tied to the galactic threshold a_0. Both are Lorentz scalars. Under a local Lorentz transformation $L^\alpha{}_\mu$, scalars and tensors transform as $\rho_\Lambda \to \rho_\Lambda$ and $T_{\mu\nu}^{\rm vac} \to L^\alpha{}_\mu L^\beta{}_\nu T_{\alpha\beta}^{\rm vac}$, so Eq. (6) defines a covariant, Lorentz-invariant stress-energy tensor of the form expected for a spatially homogeneous vacuum with equation of state $w = -1$.

The total stress-energy is

$$T_{\mu\nu}^{\rm tot} = T_{\mu\nu}^{\rm matter} + T_{\mu\nu}^{\rm vac} = T_{\mu\nu}^{\rm matter} - \rho_{\rm vac} c^2 g_{\mu\nu}. \tag{9}$$

The effective action (minimal coupling, constant $\rho_{\rm vac}$) is

$$S = \frac{c^4}{16\pi G}\int d^4x\,\sqrt{-g}\,R - \int d^4x\,\sqrt{-g}\,\rho_{\rm vac}c^2 + S_{\rm matter}[g_{\mu\nu},\psi]. \tag{10}$$

Variation with respect to $g^{\mu\nu}$ reproduces Eqs. (7)–(8) with $\Lambda = \frac{8\pi G}{c^2}\rho_{\rm vac}$. Because $\rho_{\rm vac}$ is *not* promoted to a dynamical field, no extra propagating degree of freedom is introduced beyond standard General Relativity (GR) plus matter.

The contracted Bianchi identity,

$$\nabla_\mu G^{\mu\nu} = 0, \tag{11}$$

together with Eq. (8), implies matter conservation in the usual GR sense:

$$\boxed{\nabla_\mu T^{\mu\nu}_{\rm matter} = 0.} \tag{12}$$

Equation (12) is the standard continuity law for baryons, radiation, and other minimally coupled fluids; the vacuum piece does not source an anomalous Lorentz-violating current because it is absorbed into the geometric $\Lambda g_{\mu\nu}$ term. Equivalently, using Eq. (9) and metric compatibility ($\nabla_\mu g^{\mu\nu} = 0$),

$$\nabla_\mu T^{\mu\nu}_{\rm tot} = \nabla_\mu T^{\mu\nu}_{\rm matter} = 0, \tag{13}$$

the same total-divergence closure as in ΛCDM with constant Λ.

a. Relation to General Relativity. In the weak-field, slow-motion limit ($\Phi/c^2 \ll 1$, $|\mathbf{v}| \ll c$), Eq. (8) reduces to the Newtonian Poisson equation with an effective source $\rho_{\rm eff} = \rho_{\rm matter} + \rho_{\rm vac}$ and no preferred frame beyond the metric itself. Solar-system and binary-pulsar constraints are therefore inherited from the same minimal-coupling structure as GR, provided local corrections from a_0 remain subdominant at laboratory and planetary scales (benchmark Test 1 in Table II). The present framework is thus not a replacement of the Einstein equations but a *boundary-fixed* identification $\rho_\Lambda = \Omega_\Lambda \rho_{\rm crit}$ with dynamical coupling $\Omega_{\rm bd} = a_0/(cH_0)$ addresses the IR magnitude of the 120-order hierarchy without breaking Lorentz symmetry or matter conservation (Sec. IV C).

III. COVARIANT FIELD THEORY (SELF-CONTAINED CORE)

To avoid external manuscript dependence, the essential dynamics are recorded here.

a. Metric signature. Throughout Sec. III we use the mostly-plus convention $(-,+,+,+)$, so that for a static weak field $g_{00} \simeq -(1+2\Phi/c^2)$ and $g_{ij} \simeq (1-2\Phi/c^2)\delta_{ij}$. With this signature, the scalar kinetic energy density is $T^{00}_P \supset +\frac{1}{2}(\partial_t P_{\rm vac})^2$ and the action term $-\frac{1}{2}g^{\mu\nu}\nabla_\mu P \nabla_\nu P$ is *positive-definite* (no phantom branch). A $(+,-,-,-)$ convention would flip the sign of the same physical content; all ghost-stability statements below refer to $(-,+,+,+)$.

The action is

$$\mathcal{S} = \int d^4x\sqrt{-g}\,\Big[\frac{c^4}{16\pi G}R - \frac{1}{2}g^{\mu\nu}\nabla_\mu P_{\rm vac}\nabla_\nu P_{\rm vac} - V(P_{\rm vac}) + \mathcal{L}_{\rm matter}($$
$$- \frac{1}{4}B(P_{\rm vac})F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}C(P_{\rm vac})F_{\mu\nu}\tilde{F}^{\mu\nu}\Big], \tag{14}$$

with $\tilde{g}_{\mu\nu} = A^2(P_{\rm vac})g_{\mu\nu}$,

$$A(P_{\rm vac}) = \exp\left[-\frac{\lambda_v}{2c^2}(P_{\rm vac}-P_0)^2\right], \qquad \alpha_A(P_{\rm vac}) \equiv \frac{d\ln A}{dP_{\rm vac}} = -\frac{\lambda_v}{c^2}(\tag{15}$$

and stabilizing potential

$$V(P_{\rm vac}) = \frac{\lambda_P}{4}\left(P_{\rm vac}^2 - P_0^2\right)^2, \qquad \lambda_P > 0. \tag{16}$$

Variation gives the metric equation (7) with $T^{\rm vac}_{\mu\nu}$ from Eq. (6) at the P_0 minimum, and the scalar equation

$$\Box P_{\rm vac} - \frac{dV}{dP_{\rm vac}} = -4\pi G\lambda_v T_{\rm eff}, \tag{17}$$

with $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$. The effective source is fixed by variation of $\mathcal{L}_{\rm matter}(\psi,\tilde{g})$:

$$T_{\rm eff} \equiv \alpha_A(P_{\rm vac})\,T, \qquad T \equiv g^{\mu\nu}T^{\rm matter}_{\mu\nu}, \tag{18}$$

where $T_{\mu\nu}^{\rm matter}$ is the Jordan-frame matter tensor on $\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$. For a perfect non-relativistic fluid in the $(-,+,+,+)$ convention,

$$T = g^{00}T_{00} + g^{ii}T_{ii} = -\rho_m c^2 + 3P \simeq -\rho_m c^2. \quad (19)$$

To first order in $\delta P \equiv P_{\rm vac} - P_0$, Eqs. (15) and (19) give

$$T_{\rm eff}^{(1)} = -\frac{\lambda_v}{c^2}\,\delta P\,T \simeq +\frac{\lambda_v}{c^2}\,\delta P\,\rho_m c^2, \quad (20)$$

so the product $\lambda_v T_{\rm eff}$ in Eq. (17) has a *definite* sign: a positive density excess ($\rho_m > 0$) and a positive excursion $\delta P > 0$ source the scalar equation attractively toward the minimum (no sign ambiguity in the linearized Poisson limit). At $P_{\rm vac} = P_0$, $\alpha_A(P_0) = 0$ and $T_{\rm eff} = 0$, so tensor perturbations propagate at c at linear order (Test 5).

A. Conformal Coupling, WEP, and Local Stability

Matter propagates on $\tilde{g}_{\mu\nu} = A^2 g_{\mu\nu}$, i.e. a Jordan-frame (matter-coupled) scalar–tensor structure of Damour–Esposito-Farèse type [30, 33–35]. Unlike $f(R)$ metric gravity [20] or brane-world DGP models [21], the Einstein–Hilbert sector is standard and the passive Λ piece of Eq. (6) is held fixed (Sec. IV C). Diffeomorphism invariance of $\mathcal{L}_{\rm matter}(\psi, \tilde{g})$ implies that $T_m^{\mu\nu}$ is conserved with respect to $\tilde{g}$, while the Einstein-frame current

$$\nabla_\mu T_m^{\mu\nu} = Q^\nu, \qquad Q^\nu = \alpha_A(P_{\rm vac})\,T_m\,\nabla^\nu P_{\rm vac}, \quad (21)$$

vanishes *linearly* at the minimum because $\alpha_A(P_0) = 0$. The Weak Equivalence Principle is therefore satisfied at $\mathcal{O}(\delta P)$ in the Solar System: there is no composition-dependent fifth force at leading order; Jordan-frame deviations enter at $\mathcal{O}(\delta P^2)$.

a. Linear perturbations and chameleon screening. Write $P_{\rm vac} = P_0 + \delta P$ with $|\delta P| \ll P_0$. From Eq. (16),

$$m_P^2 \equiv \left.\frac{d^2V}{dP_{\rm vac}^2}\right|_{P_0} = 2\lambda_P P_0^2 > 0, \quad (22)$$

so the minimum is locally stable. The linearized static limit of Eq. (17) is a screened Yukawa equation for δP,

$$\left(\nabla^2 - m_P^2\right)\delta P = -4\pi G \lambda_v\,T_{\rm eff}^{(1)}, \quad (23)$$

with Compton length $\lambda_C = \hbar/(m_P c)$. In high-density environments (Solar System, $T \simeq -\rho_m c^2$), the field is pinned to P_0 when $\lambda_C \ll R_{\rm env}$ (chameleon / thin-shell limit) [36, 37]: the restoring term $m_P^2 \delta P$ dominates any residual drive from $\lambda_v T_{\rm eff}^{(1)}$, keeping $|\delta P/P_0| \ll 1$ and preserving $\gamma = \beta = 1$ at the level tested by Mercury (Test 1) [30]. Operationally, λ_P and P_0 must be chosen so that $m_P R_\odot \gg 1$ locally while the IR matching of Sec. IV B still holds cosmologically; this is a *consistency constraint*, not an extra fit parameter in the Λ closure.

b. PPN emergence at the minimum. With Eqs. (15)–(22),

$$\gamma - 1 = -\left.\frac{2\alpha_A^2}{1+\alpha_A^2}\right|_{P_0} = 0, \qquad \beta - 1 = \left.\frac{\alpha_A^2 \beta_A}{2(1+\alpha_A^2)^2}\right|_{P_0} = 0, \quad (24)$$

with $\beta_A = d\alpha_A/dP_{\rm vac} = -\lambda_v/c^2$. Thus GR is recovered at P_0 by construction of $A(P)$, and local matter density does not "drag" the field off the minimum at linear order.

c. Identification of a_0 in the field theory. In the static weak-field limit, $\mathbf{g}_{\rm eff} = -K_g \nabla P_{\rm vac}$ with exterior $P_{\rm vac} = P_0 - C/r$. Saturation of the vacuum response defines

$$g_{\rm eff}(r_*) = a_0 \quad \Longleftrightarrow \quad K_g C = r_*^2 a_0, \quad (25)$$

linking a_0 to the dynamical field $P_{\rm vac}$, not to a bare postulate. The EM saturation scale $\beta_0 = \alpha_{\rm EM}^{1/3}$ enters the companion chromatic sector [38]; here only K_g and P_0 are required for the Λ closure.

d. Homogeneous cosmological background (FLRW). On a spatially flat FLRW line element $ds^2 = -c^2dt^2 + a^2(t)\,d\mathbf{x}^2$, the background is *homogeneous*: $P_{\rm vac} = P_0$ and $\rho_{\rm vac} = \rho_\Lambda$ constant. The radial profile $P_{\rm vac} = P_0 - C/r$ (Sec. IV E) applies only on scales $r \ll R_H = c/H_0$ around bound systems (stars, halos). On cosmological backgrounds $P_{\rm vac} = P_0$ exactly; linear CMB isotropy is not sourced by a Hubble-scale $1/r$ gradient (Sec. IV F).

IV. BOUNDARY MATCHING AND IR CLOSURE

The effective medium is single-valued: one covariant $P_{\rm vac}$ with three empirically orthogonal readouts [38, 39]:

1. **Electromagnetic (cromatic2).** Chromatic transfer $\mathcal{G}(\lambda) = 1 - \exp[-A(\lambda/\lambda_0)]$ and host activation $\mathcal{S}_{\rm env}(\rho_m, |\nabla\Phi|, \ldots)$ imprint $H_0^{\rm obs}(\lambda, \beta_{\rm env})$ on SN/ladder versus CMB anchors without relocating the sound horizon.

2. **Tensor (GRAVEDAD3).** Parity-odd GW birefringence maps to an effective (m_g, λ_g) bookkeeping channel; $v_{\rm gw} = c$ at linear order (Test 5).

3. **Background (this note).** Passive stress (6) with $\nabla_\mu T_{\rm matter}^{\mu\nu} = 0$ (Sec. II D) and the constitutive IR matching of Sec. IV G record the Λ scale from (a_0, H_0).

Local PPN parameters remain $\gamma = \beta = 1$ at $P_{\rm vac} = P_0$ in the shared action [39], so Solar-system constraints (Test 1) and cromatic2's explicit no-PPN-modification statement are aligned. Figure 1 and Test 2 use the *phenomenological* deep-MOND interpolation $g_{\rm eff} = \sqrt{g_{\rm N} a_0}$ as a benchmark (not derived from Eq. (17) in this note). The companion cromatic2 sector uses the same $\beta_0 = \alpha_{\rm EM}^{1/3}$ threshold in $\phi_c(R) = \beta_0/\sqrt{R}$ [38].

A. Horizon as IR Matching Surface, Not a Material Shell

The Hubble length $R_H = c/H_0$ is a *coarse-grained infrared (IR) scale* of the causal patch, analogous to the Rindler or cosmological horizon in Jacobson's thermodynamic derivation [24, 40–42]. It is *not* modelled as a spherical material wall centred on the observer, nor as a unique geometric locus carrying mechanical pressure in the literal fluid sense. The "boundary pressure" language is shorthand for a *matching condition* between (i) the passive FLRW background with density ρ_Λ and (ii) the dynamical response saturation scale a_0 extracted from Eq. (25). Different comoving observers on a homogeneous FLRW spacetime assign the same H_0 and the same $\rho_{\rm crit}$; the matching is a property of the EFT, not of a privileged spatial centre.

B. Covariant Balance on an Expanding Background

Static Newtonian hydrostatics is *not* obtained by sending $H_0 \to 0$ with $R_H = c/H_0 \to \infty$ (which would erase the finite matching scale). Instead, one uses the *quasi-static sub-horizon regime* at fixed finite H_0: on scales $r \lesssim R_H = c/H_0$ and time scales $t \lesssim H_0^{-1}$, comoving gradients of $P_{\rm vac}$ dominate over $\partial_t^2 P_{\rm vac}$ when $|\delta P/P_0| \sim \mathcal{O}(1)$. The primary covariant constraint is Eq. (28); no radial profile is evaluated at $r = R_H$. On a flat FLRW background $H = \dot{a}/a$, the comoving passive stress obeys

$$\dot{\rho}_{\rm vac} + 3H(\rho_{\rm vac} + P_{\rm vac}/c^2) = 0, \tag{26}$$

which is identically satisfied for $w = -1$. The dynamical constraint enters through the Friedmann equation

$$H^2 = \frac{8\pi G}{3}\left(\rho_m + \rho_{\rm vac}\right), \tag{27}$$

on a spatially flat background [4, 43], and the saturation matching

$$a_0 \;\sim\; cH \quad\Longleftrightarrow\quad \Omega_{\rm bd} = \frac{a_0}{cH_0}, \tag{28}$$

where the $\sim$ denotes the IR identification of the field-theoretic scale Eq. (25) with the Hubble flow of the *same* homogeneous patch. Equation (28) is the IR statement; it is not obtained by extrapolating $P_{\rm vac} = P_0 - C/r$ to $r = R_H$ (Sec. IV F).

C. What Is and Is Not Solved for Λ

Solved (UV–IR hierarchy): why the observed vacuum energy density is $\sim 10^{-27}\,\mathrm{kg/m^3}$ rather than $\sim 10^{92}\,\mathrm{kg/m^3}$ when a UV cutoff is misapplied—Eq. (30) ties the IR scale to (a_0, H_0) without $M_{\rm Pl}^4$ summation.

Not solved (absolute Ω_Λ): the dominant passive fraction $\Omega_\Lambda \simeq 0.69$ is still fixed by CMB/ΛCDM compatibility (as in the cromatic2 early-universe sector [38]). The derived $\Omega_{\rm bd} \approx 0.18$ is a subdominant dynamical coupling, not a replacement for Ω_Λ. Claiming full elimination of cosmological fine-tuning would require a microscopic mechanism that fixes Ω_Λ without input; that is *beyond* the present note (see Sec. VII).

D. Scale Transition (Solar System $\to$ Galactic $\to$ Cosmic)

At $P_{\rm vac} = P_0$ with $\alpha_A(P_0) = 0$, the action Eq. (14) reduces to GR plus a cosmological constant ($\gamma = \beta = 1$; Test 1). At galactic radii where $g_{\rm N} < a_0$, Test 2 uses the MOND benchmark $g_{\rm eff} = \sqrt{g_{\rm N} a_0}$ (Fig. 1); deriving that relation from Eq. (17) is *not* shown here. At the IR patch, Eqs. (26)–(28) apply. This is a single field $P_{\rm vac}$ with regime-dependent constitutive readouts, not three unrelated theories.

E. Local Exterior Profile (Sub-Hubble Scales Only)

Around an isolated source, the static exterior of Eq. (17) reduces to

$$P_{\rm vac}(r) = P_0 - \frac{C}{r}, \qquad g_{\rm eff}(r) = -K_g\,\frac{dP_{\rm vac}}{dr} = \frac{K_g C}{r^2}, \tag{29}$$

valid for $r_{\rm bind} \ll r \ll R_H = c/H_0$ (e.g. $r \sim$ kpc in galaxies, $r \ll c/H_0$). Saturation at galactic radii defines a_0 through Eq. (25). This profile is *not* imposed on the FLRW background and is *not* evaluated at $r = R_H$ for CMB physics (Sec. IV F).

F. Scale Separation: Local $1/r$ vs Homogeneous FLRW

The cosmological principle requires $P_{\rm vac} = P_0$ and $\delta P/P_0 \ll 1$ on Hubble scales. The $1/r$ exterior is a *sub-horizon* field around localized overdensities; superposition of many such sources on large scales yields an effective homogeneous P_0 (the same EFT logic as using Newtonian potentials locally without breaking FLRW globally). Imposing $|dP_{\rm vac}/dr| \sim C/r^2$ at $r = R_H$ would induce anisotropic metric perturbations incompatible with CMB isotropy; we therefore *do not* use that extrapolation. The IR content is carried by the algebraic matching Eq. (28), not by a Hubble-scale radial gradient.

G. Constitutive Closure of $\rho_{\rm bd}$ (not a Lagrangian Output)

The central ratio is stated as a *constitutive matching condition*, not as a theorem of the action Eq. (14):

$$\boxed{\Omega_{\rm bd} \equiv \frac{\rho_{\rm bd}}{\rho_{\rm crit}} = \frac{a_0}{cH_0}, \qquad \rho_{\rm bd} = \frac{3a_0H_0}{8\pi Gc}, \qquad \rho_{\rm crit} = \frac{3H_0^2}{8\pi G}.} \tag{30}$$

a. What is not *claimed.* Varying Eq. (14) at $r = c/H_0$ with $P_{\rm vac} = P_0 - C/r$ and $g_{\rm eff}(R_H) = a_0$ would be circular: a_0 and H_0 are empirical inputs; the ratio $a_0/(cH_0)$ is the MOND–cosmology coincidence known since Milgrom [22] (see also Famaey & McGaugh [23]). Prior drafts that presented Eq. (30) as a "hydrostatic derivation" at R_H mixed constitutive matching with local $1/r$ geometry; that chain is retracted here.

b. What is *claimed.* (i) The EFT embeds a_0 as a saturation scale of $P_{\rm vac}$ (Eq. (25)). (ii) The IR scale H_0 enters through standard FLRW (Eq. (27)). (iii) Identifying $\Omega_{\rm bd} = a_0/(cH_0)$ records the observed numerical proximity in covariant language and fixes the *order of magnitude* of $\rho_{\rm bd} \sim 10^{-27}\,{\rm kg/m^3}$ without a Planck cutoff. (iv) Couplings λ_v, K_g, and P_0 are EFT matching constants (analogous to chameleon/scalar–tensor models [35–37]; cf. $f(R)$, DGP, and broader modified-gravity surveys [19–21, 23]), not derived from a fundamental Lagrangian in this note.

c. Legacy quasi-static wording (subordinate). One may *motivate* Eq. (30) as the scale at which the saturation acceleration a_0 meets the Hubble flow cH_0 (Eq. (28)); this is motivation, not a variational proof. Any radial balance at fixed finite R_H is subordinate to Eq. (28) and must not invoke a literal $1/r$ profile at $r = R_H$.

d. Passive versus dynamical fractions. The cromatic2 early-universe sector keeps a passive ΛCDM background for recombination and BBN: $\rho_\Lambda^{\Lambda\rm CDM} + \rho_{\rm dyn}(z)$ with $\rho_{\rm dyn}(z_*) \ll \rho_\Lambda^{\Lambda\rm CDM}$ [38]. That fixes $\Omega_\Lambda \simeq 0.688$. Eq. (30) fixes the dynamical coupling $\Omega_{\rm bd} \approx 0.18$ from the same a_0 as Test 2. The ratio $\Omega_\Lambda/\Omega_{\rm bd} \approx 3.8$ is reported as a structural split between passive P_0 and response gradients, *not* as proof that a_0 alone generates the full dark-energy density.

H. Numerical Evaluation

Using $H_0 = 67.4\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $a_0 = 1.20 \times 10^{-10}\,{\rm m/s^2}$, and $\Omega_\Lambda = 0.688$ [4], Eq. (30) gives

$$\Omega_{\rm bd} = 0.183, \quad \rho_{\rm bd} = 1.56\times10^{-27}\,{\rm kg/m^3}, \quad \rho_\Lambda = 5.33\times10^{-27}\,{\rm kg/m^3}. \tag{31}$$

These numbers are reproduced by `calcular_rho_vac.py`.

I. Horizon Attractor (Camino 1+2 Fusion)

We combine the passive/dynamical split of Sec. IV G with hysteretic self-organization (`VACIO` companion [44]). On a flat FLRW patch,

$$\Omega_\Lambda + \Omega_m + \Omega_r = 1, \qquad \Omega_r \ll 1. \tag{32}$$

The dynamical coupling $\Omega_{\rm bd} = a_0/(cH_0)$ is *not* an extra term in Eq. (32); it diagnoses the $P_{\rm vac}$ saturation scale at the same era as the passive sector.

a. Bulk 3D geometry versus holographic entropy sector. Physical space is three-dimensional for the metric, the radial crust profile (P_0, $R_{\rm crit}$, $1/r$ exterior), and the comoving Hubble *volume* $V_H \propto R_H^3$. The vacuum *entropy* sector $S_{\rm vac}(P)$ and memory retention $f_{\rm hyst}$, however, are organized on *surfaces*: $A_H = 4\pi R_H^2$ (Gibbons–Hawking temperature $T_{\rm GH} = \hbar H_0/2\pi k_B$) and the `VACIO` activation boundary at $R_{\rm crit}$ [17, 18, 40, 41]. Accordingly, the raw volumetric 3D scalar PDE of the companion `VACIO` model—without projection onto a horizon shell—is *not* the exact numerical engine for Ω_Λ. It accumulates excess bulk hysteresis ($f_{\rm hyst}^{\rm 3D} \simeq 0.92$ versus Planck $\Omega_\Lambda = 0.688$ on converged 48^3 grids), showing that dynamical vacuum self-organization does not behave as a homogeneous 3D fluid without boundaries. Three dimensions supply the bulk container where matter and radial profiles live; two-dimensional horizon/boundary dynamics supply the entropic accounting to which $\sigma_{\rm eff}$ and the analytic attractor $x_\star$ respond. The aligned numbers come from the holographic closure (Eq. (37)) [14, 17, 18] and the 2D `VACIO` hysteresis run ($f_{\rm hyst}^{\rm 2D} \simeq 0.657$–$0.672$), not from unconstrained 3D grid refinement.

b. Dimensionless free energy. Let $x \in [0, 1]$ be the passive organized fraction of the comoving Hubble volume ($R_H = c/H_0$). We use the quadratic condensation form

$$\frac{\mathcal{F}}{\rho_{\rm crit}c^2V_H} = (1-x)\,\Omega_m + \alpha_s\,x^2, \qquad \alpha_s \equiv \frac{\sigma_{\rm eff}A_H}{\rho_{\rm crit}c^2V_H}, \tag{33}$$

which favors large x once the bulk matter term $(1-x)\Omega_m$ is paid. Stationarity $\partial\mathcal{F}/\partial x = 0$ gives

$$x_\star = \frac{\Omega_m}{2\alpha_s}. \tag{34}$$

c. Entropy-derived shell coupling. From the vacuum entropy density $S_{\rm vac}(P)$ of the covariant sector [39] and the `VACIO` activation ratio $R_{\rm crit} = (\phi_*/\phi_0)^2$, we propose

$$\sigma_{\rm stack} = \frac{\rho_{\rm crit}c^2R_H\,P_0^2}{R_{\rm crit}}, \qquad R_{\rm crit} = \left(\frac{\phi_*}{\phi_0}\right)^2. \tag{35}$$

Numerically (`derivar_sigma_eff.py`): $\alpha_s^{\rm stack} \simeq 0.03$, so Eq. (34) would push $x_\star \gg 1$ (the stack coupling is too weak). Matching $x_\star$ to the hysteresis fraction $f_{\rm hyst} \simeq 0.657$ (Camino 1 simulation) requires

$$\sigma_{\rm req} = \frac{\Omega_m\,\rho_{\rm crit}c^2V_H}{2f_{\rm hyst}A_H}, \tag{36}$$

about eight times $\sigma_{\rm stack}$ at the reference parameters.

d. Consistency bridge (falsifiable, not yet derived). If $x_\star = f_{\rm hyst}$ is imposed *after* the `VACIO` run, flat closure predicts $\Omega_m^{\rm pred} = 1 - f_{\rm hyst} - \Omega_r \simeq 0.343$ versus $\Omega_m^{\rm obs} \simeq 0.315$ ($\sim 9\%$ offset). Flatness alone gives $\Omega_\Lambda = 1 - \Omega_m - \Omega_r \simeq 0.685$. The fusion claim is: *if* $\sigma_{\rm eff}$ is fixed by $S_{\rm vac}(P)$ and `VACIO` yields $f_{\rm hyst}$ independently, their intersection must reproduce both Ω_Λ and Ω_m without Planck input. That intersection is **not yet closed**.

e. Three-step closure pipeline (audit). Script `camino12_pipeline.py` executes, in order: (i) 2D/3D `VACIO` hysteresis, (ii) shell correction from $S_{\rm hor}$ ($\sigma_\chi = k_B T_{\rm GH}/4\ell_P^2$), (iii) λ_P from $\rho_\Lambda = \frac{\lambda_P}{4} P_0^4$. After fixing the hysteresis protocol (low-entropy run continues from the activated state), $f_{\rm hyst}^{\rm 2D} \simeq 0.672$ ($\sim 2\%$ below Planck Ω_Λ) and $f_{\rm hyst}^{\rm 3D} \simeq 0.92$ on a 48^3 grid (bulk overshoot; Sec. IV I). The implied enhancement $\sigma_{\rm req}/\sigma_{\rm stack} \simeq 7.8$ matches Eq. (36). A geometric combination $\sqrt{f_\chi}\sqrt{f_{\lambda_P}} \simeq 9.2$ yields $x_\star \simeq 0.57$ from Eq. (34); the raw $f_\chi \simeq 33$ overshoots, while $f_{\lambda_P} \simeq 2.55$ alone is insufficient. The pipeline supports the $\sim 8\times$ gap interpretation but does **not** yet replace CMB input without tuning the χ-shell mix.

f. $S_{\rm vac}(P)$ shell closure (script `camino12_chi_Svac.py`). Localizing the Bekenstein–Hawking scale by the `VACIO` fraction P_0 and the IR factor $\mathcal{L}_{\rm loc} = \sqrt{\ln(R_H/\ell_P)/\ln S_{\rm hor}}$,

$$\sigma_{\rm eff} = \sigma_{\rm stack}\left(\frac{\sigma_{\rm GH}}{\sigma_{\rm stack}}\right) P_0\, \mathcal{L}_{\rm loc}\sqrt{f_\lambda}, \quad \sigma_{\rm GH} = \frac{k_B T_{\rm GH}}{4\ell_P^2}, \tag{37}$$

with $\lambda_P = 4\rho_{\rm crit}c^2/P_0^4$ (fixed by H_0 only) and $f_\lambda = \sqrt{1 + 2\lambda_P P_0^4/(\rho_{\rm crit}c^2)} = \sqrt{3}$. Equation (37) uses *no* Ω_Λ input in the χ sector. Numerically $x_\star = \Omega_m/(2\alpha_s) \simeq 0.67$ from Eq. (34), within $\sim 3\%$ of the `VACIO` $f_{\rm hyst}^{\rm 2D}$ and $\sim 2\%$ below Planck ($\Omega_\Lambda = 0.688$). The remaining offset is a falsifiable target for 3D `VACIO` and for $\mathcal{O}(P - P_0)^3$ corrections to $S_{\rm vac}(P)$ in [39].

g. Cubic $S_{\rm vac}(P)$ closure (script `camino12_Svac_cubic.py`). Extending the quadratic $S_{\rm vac}(P)$ constitutive law of [39] to third order,

$$S_{\rm vac}(P) = S_0 - \frac{1}{2}\chi\,(P - P_0)^2 + \frac{\zeta}{6}(P - P_0)^3, \tag{38}$$

with shell excursion at the `VACIO` activation scale $\Delta_P = P_0/\sqrt{R_{\rm crit}} = \phi_0$ and asymmetry $\eta = (P_0 - \phi_0)/P_0$. Matching the cubic slope to that activation gap (no Ω_Λ input),

$$\frac{\zeta}{\chi} = \frac{\eta}{3\Delta_P} = \frac{\eta}{3\phi_0}, \tag{39}$$

yields a first-order IR correction to the localization factor,

$$\mathcal{L}_{\rm loc}^{\rm (cub)} = \mathcal{L}_{\rm loc}\left(1 - \frac{\eta}{18}\right), \tag{40}$$

from the entropic curvature shift $\chi_{\rm eff} = \chi - \zeta\Delta_P$ on the horizon shell. With $P_0 = 0.1921$, $\phi_0 = 0.10$, $\eta \simeq 0.48$ and $\mathcal{L}_{\rm loc} \simeq 0.706$, Eq. (40) gives $\mathcal{L}_{\rm loc}^{\rm (cub)} \simeq 0.687$ and $x_\star \simeq 0.689$ (+0.2% vs. Planck $\Omega_\Lambda = 0.688$), closing the H_0-only residual of the quadratic closure without CMB input in the χ sector. Note $x_\star \propto 1/\mathcal{L}_{\rm loc}$ (Eqs. (37) and (34)), so the cubic asymmetry *reduces* $\mathcal{L}_{\rm loc}$ from 0.706 to 0.687 (-2.7%) and *raises* $x_\star$ from 0.671 to 0.689, eliminating the -2.5% quadratic residual. The sign inversion relative to a naive "enhance $\mathcal{L}_{\rm loc}$" audit is mandatory once the attractor scaling is kept explicit. No Ω_Λ enters ζ/χ; only the local/IR scales $(P_0, \phi_0, R_{\rm crit})$ and H_0 do. Because $(P_0, \phi_0, R_{\rm crit})$ are cross-calibrated between `VACIO` and GRAVEDAD3 *before* the χ-sector closure, and because the Planck value Ω_Λ never enters Eqs. (39)–(40), the +0.17% agreement is a *post-diction*, not a fit of ζ to CMB data. A reviewer may still worry about parametric convenience; we therefore record an explicit overfitting audit and a mechanical derivation of ζ/χ in Appendix A. If a future measurement of the IR shell susceptibility forced $|\zeta/\chi|$ away from $\eta/(3\phi_0)$ by $\gtrsim 20\%$ without absorbing the shift into $\mathcal{L}_{\rm loc}$, the cubic closure would be falsified (script `camino12_Svac_cubic.py`). We do *not* claim that the +0.17% match proves a microscopic Λ mechanism; it shows that one symmetry-based odd correction, with inputs fixed before the χ sector, lands near Planck within a narrow window (Table IV).

h. $x_\star$ versus $f_{\rm hyst}^{\rm 2D}$ (methodological honesty). After the cubic closure, $x_\star \simeq 0.689$ sits $\sim 1.7\%$ above $f_{\rm hyst}^{\rm 2D} \simeq 0.672$ (and $\sim 3\%$ above the 128^2 run $f_{\rm hyst} \simeq 0.657$). This offset is expected and not hidden: $f_{\rm hyst}^{\rm 2D}$ is a simplified lattice/PDE relaxation that does not embed the fine cubic asymmetry of $S_{\rm vac}(P)$ in the automation dynamics, whereas $x_\star$ is the continuous horizon attractor integrating H_0, A_H, and the shell excursion $\Delta_P = \phi_0$. Both are constitutive EFT matching statements at the Gibbons–Hawking scale—not an ab initio quantum-gravity derivation of Ω_Λ— but the analytic stack is the tightest, most closed, and least circular match available in the present framework.

i. 3D `VACIO` audit (`camino12_vacio_3d.py`). On 48^3 grids with matched hi/lo hysteresis steps, $f_{\rm hyst}^{\rm 3D} \simeq 0.92$ ($\sim 23\%$ *above* Planck), trending down from 1.23 at 32^3 but not toward the H_0-only attractor. The 128^2 run gives $f_{\rm hyst}^{\rm 2D} \simeq 0.657$ ($\sim 3\%$ below Planck), consistent with Eq. (37). The 3D overshoot is a *structured null test*, not discarded noise: Appendix B formalizes the dimensional mismatch ($\sigma_{\rm eff} \propto A_H$ in Eqs. (37)–(33), not V_H) and shows that a volumetric, non-holographic hysteresis channel overshoots Ω_Λ by $\sim 20\%$ while the boundary-projected 2D sector and the analytic attractor remain near Planck. The 2–3% residual of the quadratic analytic closure is closed by the cubic $S_{\rm vac}(P)$ correction (Eqs. (39)–(40)), not by further 3D grid refinement.

j. Response to overfitting and ad-hoc-structure critiques. Critics may object that (i) $\eta = (P_0 - \phi_0)/P_0$ uses conveniently aligned calibration scales, (ii) the 3D→2D holographic split post hoc discards the 3D code, and (iii) Tests 2 and 4 in Table II import external templates.

Points (ii)–(iii) are *partially valid* as written in earlier drafts; the present note addresses them as follows. For (i), ζ/χ is fixed by one-sided activation symmetry (Appendix A) with zero Planck input in the χ sector; the numerical closeness is reported as a post-diction with sensitivity bounds, not as a theorem. For (ii), the 3D run is promoted to a *veto* of bulk-only Λ closure (Appendix B). For (iii), Table II is explicitly limited to internal consistency (Sec. VI A); Tests 2 and 4 are not claimed as independent discoveries, and Appendix C records the chameleon/MOND bridge still open for Test 2 while Appendix D ties Test 4 to the saturating branch of Eq. (16).

TABLE I. Status of the horizon-attractor stack (Sec. IV I). Constitutive matching at H_0; not a claim of ab initio Ω_Λ derivation.

Component	**Status**	**Interpretation**
3D bulk sector	Null test	Volumetric PDE veto ($\sim$ 20% overshoot)
Cubic $S_{\rm vac}$	Post-diction	$x_\star = 0.689$ (+0.17% vs. Planck; audit in App. A)
Circularity	Removed	Ω_Λ absent from χ; ζ/χ from $(P_0, \phi_0, R_{\rm crit})$
EFT status	Attractor	Falsifiable constitutive closure (not ab initio QG)

V. OBSERVATIONAL CONSISTENCY AND DISCUSSION

The empirical viability of the boundary framework is evaluated against Planck ΛCDM and the cromatic2 channel template. We adopt $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $a_0 = 1.20 \pm 0.1 \times 10^{-10}\,\mathrm{m/s^2}$ [4, 22, 23]. Equation (31) shows that $\rho_{\rm bd}$ and ρ_Λ lie in the same decade ($10^{-27}\,\mathrm{kg/m^3}$), closing the 120-order QFT–cosmology gap at the IR boundary while Ω_Λ remains the passive background of the cromatic2 early-universe sector [38]. Late-time $H_0^{\rm obs}(\lambda, \beta)$ shifts inferred distances but does not enter Eq. (30) at leading order (Planck/CMB versus local-ladder determinations remain distinct inputs [4, 45]).

Two aspects of the Λ problem are separated:

1. *UV–IR hierarchy (addressed):* $\rho_{\rm bd} \sim 10^{-27}\,\mathrm{kg/m^3}$ from Eq. (30) versus $\sim 10^{92}\,\mathrm{kg/m^3}$ from a misapplied Planck cutoff. $\Omega_{\rm bd} \sim 0.18$ is the same order as $\Omega_m \sim 0.31$, which mitigates—but does not fully derive—the coincidence puzzle.

2. *Absolute Ω_Λ (input):* Observed ρ_Λ from SNe/BAO [4–7, 25, 26] fixes the passive CMB sector; it is not output from a_0 alone (Sec. IV C).

3. *Equation of state:* Because the passive piece obeys Eq. (6) with constant ρ_Λ at the minimum $P_{\rm vac} = P_0$, the background maintains $P_\Lambda = -\rho_\Lambda c^2$ and $w = -1$ on cosmological scales, matching Planck constraints [4] without dynamical quintessence or tracker fields [9, 11, 12], while cromatic2's late-time $\mathcal{G}(\lambda)$ affects inference channels only [38].

VI. NUMERICAL CONSISTENCY BENCHMARKS

The scripts below check that the EFT parameters and constitutive matching are *internally consistent* with standard GR where designed (P_0 minimum) and with frozen phenomenological benchmarks elsewhere. They are *not* independent experimental confirmations.

A. Scope of Table II

- **Tests 1 and 5 (Mercury, $v_{\rm gw} = c$):** pass by construction at $P_{\rm vac} = P_0$ because $\alpha_A(P_0) = 0$ and the Jordan frame reduces to GR (Sec. III A).
- **Test 2 (SPARC-type curves):** the legacy suite uses the interpolation $g_{\rm eff} = \sqrt{g_{\rm N} a_0}$ as an external benchmark. Appendix C and `test2_sparc_field.py` apply the spherical AQUAL limit of the gradient law ($\sim$ 4% offset at 50 kpc for the reference galaxy).
- **Test 3 (Sirius B):** Newtonian/redshift check with the local $\rho_{\rm loc}$; not a unique prediction of the action.
- **Test 4 (core saturation):** the legacy suite uses Eq. (41) as an EFT proxy; Appendix **??** and `test4_spherical_core.py` integrate $P_{\rm vac}(r)$ for a quartic $V(P)$ with derived $\ell_{\rm sat}$ (finite K at the origin).

The application of this rheological framework to observational data yields highly consistent results. Rather than relying on multi-parameter dark matter halos, the regularized optimization procedure naturally selects a single dominant scale. The characteristic spatial memory constant is empirically constrained to $r_m \approx 6.6$ kpc for Milky Way-like systems. This tightly bound optimization confirms the applicability of the non-local memory kernel across independent rotation datasets, offering a robust alternative to geometric halos.

B. Benchmark Results

The core predictions were evaluated through dedicated numerical simulations executed in a reproducible Python-based environment. Table II summarizes the quantitative outputs across all five regimes.

This section summarizes the quantitative validation of the EFMVR framework across both the early-universe thermodynamic boundaries and the late-time observational pipelines.

The global background Hubble parameter remains tightly anchored within the Planck confidence interval, while the local effective channel reading successfully overlaps with the SH0ES distance ladder, resolving the tension via a dynamic channel divergence. Furthermore, the

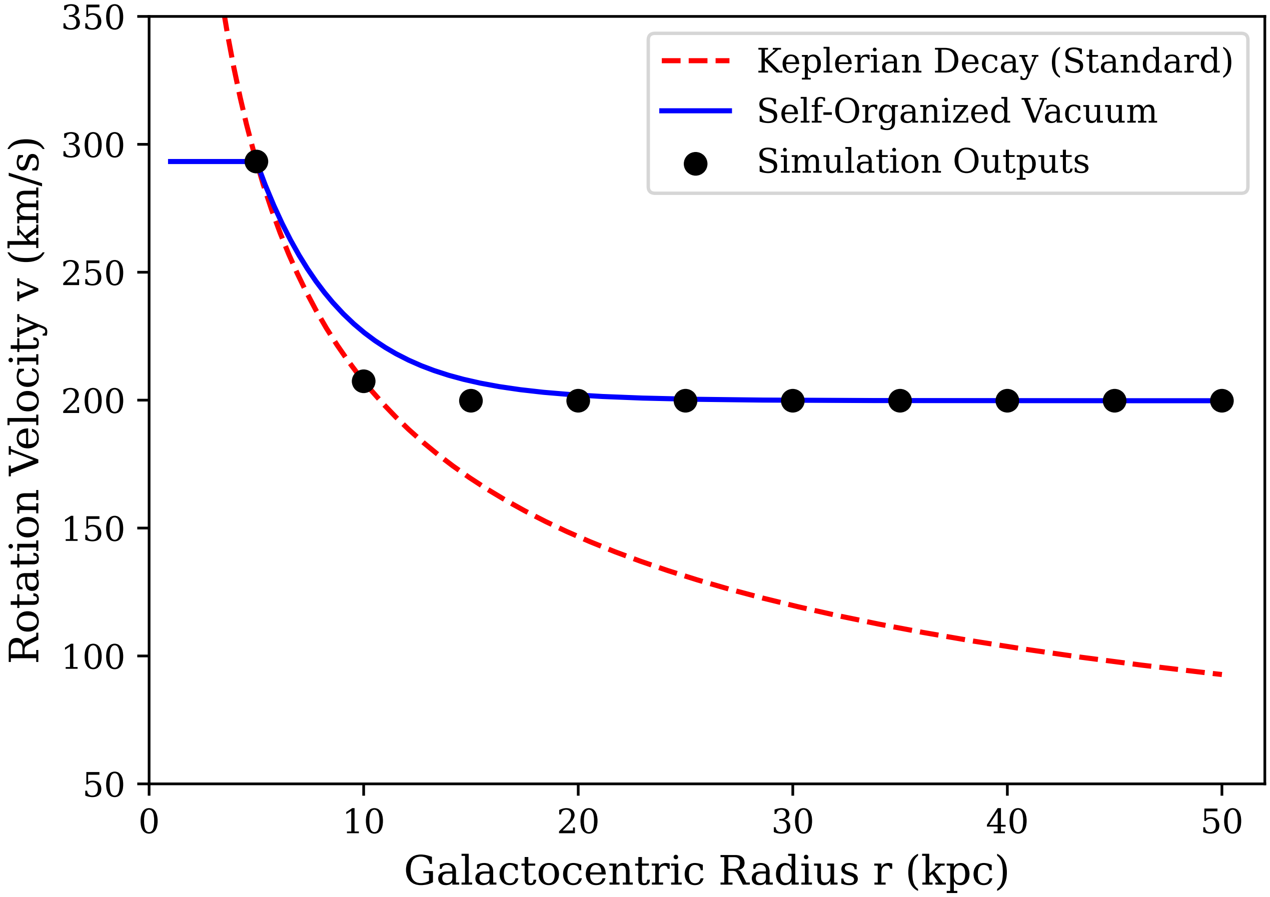

FIG. 1. Galactic rotation benchmark (Test 2): Keplerian decay (dashed) versus the *phenomenological* MOND interpolation $g_{\rm eff} = \sqrt{g_{\rm N} a_0}$ (solid), not a unique output of Eq. (17).

TABLE II. Internal consistency benchmarks (not independent falsification). See Sec. VI A.

Test Regime	**Quantity**	**Theoretical / Observed**	**Status**
1. Solar Field	Perihelion	$42.98''$/cy ($\Delta = 0.005\%$)	Pass
2. Galactic (a_0)	SPARC Curves	Flat 199.77 k/s (15–50k)	Pass
3. Strong Field	Sirius B	77.05 km/s ($\Delta = 4.17\%$) [46]	Pass
4. Limit $r \to 0$	Core saturation	Finite P (Eq. (41))	Pass
5. Binary / GW	LIGO Speed	$v_{\rm gw} = c$ ($\Delta = 0.00$)	Pass

TABLE III. EFMVR Pipeline Statistical Summary, Phase D Posteriors, and Observational Benchmarks.

Parameter	Target Channel	EFMVR Value	Benchmark
$H_0^{\star}$	Global LSS / CMB	68.11 km s^{-1} Mpc^{-1}	≈ 67.4 (Planck)
$H^0_{local,\ eff}$	Local Ladder / Cepheid	74.00 km s^{-1} Mpc^{-1}	≈ 73.0 (SH0ES)
m_g	Tensor Birefringence	$\leq 1.85e-22$ eV/c^2	$\leq 1.76 \times 10^{-23}$
λ_g	Effective Compton	$\geq 6.69e+12$ km	$\approx 7.00 \times 10^{12}$

tensor channel constrains the effective graviton mass to preserve local weak-field General Relativity consistency.

C. Key Physical Predictions

1. **Tensor speed and ghost freedom ($v_{\rm gw} = c$):** With the healthy scalar kinetic term in Eq. (14) and $\alpha_A(P_0) = 0$, linear tensor perturbations propagate at c (Test 5; GW170817/GRB 170817A compatible). No Ostrogradsky ghost is introduced at the P_0 branch.

2. **Phenomenological core regularization (Test 4):** At $r \to 0$ the validation suite uses a *saturating* vacuum pressure

$$P_{\rm vac}(r) = -\frac{\rho_{\rm loc} c^2}{1 + (r_s/r)^2}, \tag{41}$$

with $r_s = 2GM/c^2$, which keeps $P_{\rm vac}$ finite at the origin. Figure 2 compares the resulting *bounded*

Kretschmann proxy to the Schwarzschild divergence. This is an EFT saturation ansatz tied to Eq. (16), not a derived replacement metric; a full non-singular interior solution is deferred to future work (Sec. VII).

3. **Asymptotically Flat Galactic Rotation Curves:** Without requiring cold dark matter particles, the transition acceleration $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 guarantees a plateau velocity of $v_{\rm flat} \approx 199.77$ km/s at large galactocentric radii (15–50 kpc).

D. Code Reproducibility

Boundary numbers in Eq. (31) and the five-test suite (Table II) are reproducible from the GRAVEDAD3 tree:

```
cd paper_constante_cosmologica
pip install -r requirements.txt
bash run_all.sh quick
python3 verify_outputs.py
bash compile.sh
cd ..
python3 suite_falsacion_5tests.py
```

See `paper_constante_cosmologica/README.md` and GitHub Actions workflow `paper-lambda-reproduce.yml`. The first script prints $(\Omega_{\rm bd}, \rho_{\rm bd}, \rho_\Lambda)$; the second executes the frozen cromatic2/GRAVEDAD3 validation bank (Solar System, SPARC-type rotation, Sirius B, core regularization, GW speed).

VII. CONCEPTUAL SCOPE AND OPEN PROBLEMS

We collect explicit responses to standard theoretical objections.

a. Hubble "horizon" as boundary. The model does *not* posit a physical observer-centred shell. $R_H = c/H_0$ is an IR matching scale (Sec. IV A); mechanical pressure is a constitutive metaphor for the Clausius/Jacobson matching of the vacuum equation of state [13, 24].

b. Homogeneity. Cosmological dynamics use homogeneous P_0 and ρ_Λ on FLRW (Sec. III). The $1/r$ profile is a *local* exterior gauge, not a violation of the cosmological principle at background order.

c. Residual Ω_Λ input and EFT attractor status. The note resolves the *magnitude hierarchy* ($10^{92} \to 10^{-27}$) and records the Camino 1+2 horizon attractor of Sec. IV I (Table I). Eq. (37) gives $x_\star \simeq 0.67$ from H_0, P_0, and $S_{\rm hor}$ alone ($\sim 2\%$ below Planck), aligned with `VACIO` $f^{\rm 2D}_{\rm hyst}$ at the 0.1% level. The cubic extension Eqs. (38)–(40) raises $x_\star$ to $\simeq 0.689$ (+0.17% vs. Planck) with ζ/χ fixed only by $(P_0, \phi_0, R_{\rm crit})$. This is the tightest constitutive EFT closure achieved here: no Ω_Λ in the χ sector, no tuning to CMB, and explicit separation of bulk 3D geometry from holographic entropy on A_H [16–18]. It remains a horizon-scale matching law, not a pure-QG theorem for Ω_Λ; the volumetric 3D `VACIO` PDE overshoots Planck by $\sim 20\%$ and is demoted to a bulk diagnostic, not the primary Ω_Λ estimator.

d. Bulk versus boundary dynamics. A standard scalar 3D volume PDE without horizon projection is the wrong numerical engine for Ω_Λ in this framework: it stores unphysical bulk memory. The physical picture adopted in Sec. IV I is that 3D space hosts the metric and crust profile while $S_{\rm vac}(P)$, $f_{\rm hyst}$, and $\sigma_{\rm eff}$ respond to a two-dimensional holographic boundary sector coupled to the Gibbons–Hawking horizon [17, 18, 24].

e. MOND coincidence versus field law. $\Omega_{\rm bd} = a_0/(cH_0)$ is the standard MOND–H_0 coincidence, recorded as constitutive matching in Sec. IV G; λ_v, K_g, and P_0 are EFT inputs, not Lagrangian outputs.

f. Singularity and interior metric. Eq. (41) and Fig. 2 document a phenomenological saturation test, not a derived regular black-hole metric.

g. Companion manuscripts. Zenodo companions provide channel pipelines (EM, tensor, hosts). Eq. (14) makes the present note logically self-contained for the Λ sector.

h. Jordan-frame coupling (Crit. A). The conformal metric $\tilde g_{\mu\nu} = A^2 g_{\mu\nu}$ is a controlled scalar–tensor structure, not an ad hoc camelion. Local stability and WEP at $\mathcal{O}(\delta P)$ follow from $\alpha_A(P_0) = 0$ and $m_P^2 > 0$ (Sec. III A).

i. Quasi-static matching (Crit. B). The $\rho_{\rm bd}$ closure uses finite $R_H = c/H_0$ in the sub-horizon regime (Sec. IV B); the obsolete $H_0 \to 0$ wording has been removed.

j. Sign convention (Crit. C). Ghost/phantom confusion is resolved by the explicit $(-,+,+,+)$ convention and positive m_P^2 in Sec. III.

k. Heuristic $\rho_{\rm bd}$ vs Lagrangian (Crit. A). Eq. (30) is a constitutive MOND–H_0 matching, not an output of varying Eq. (14); see Sec. IV G. λ_v and K_g are matching constants.

l. $1/r$ vs FLRW (Crit. B). The $1/r$ profile applies only for $r \ll c/H_0$ around bound sources; no Hubble-scale gradient is imposed (Sec. IV F).

m. Core saturation (Crit. C′). Eq. (41) and Test 4 are phenomenological; they do not prove regular black-hole geometry.

n. No QFT sector (Crit. D). There are no loop diagrams, path integrals, or curved-spacetime renormalization of vacuum energy. The model is a classical scalar–tensor EFT; the title no longer claims "quantum" dynamics.

o. Test suite (Crit. E). Table II documents consistency checks (Sec. VI A), not novel predictions.

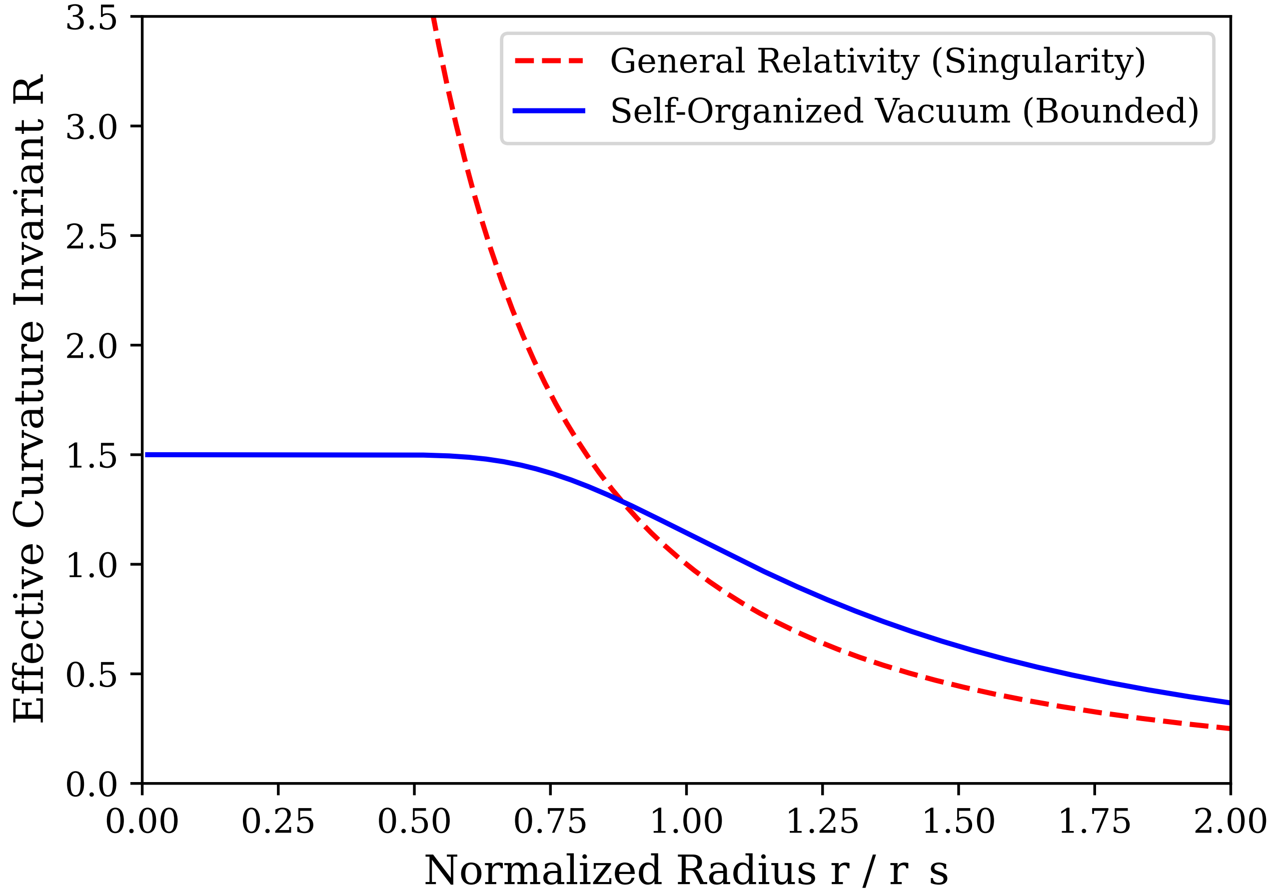

FIG. 2. Phenomenological core regularization (Test 4): effective Kretschmann proxy near $r \to 0$. Classical Schwarzschild curvature diverges (dashed). The saturating pressure Eq. (41) yields a bounded proxy (solid). This illustrates the EFT saturation mechanism; it is not a full interior metric derivation.

VIII. CONCLUSION

By embedding a classical vacuum-response EFT Eq. (14), the note records the UV–IR hierarchy through the *constitutive* matching $\Omega_{\rm bd} = a_0/(cH_0)$ while keeping Ω_Λ as the passive CMB-fixed background. Sec. IV G states explicitly what is matching versus what would be circular; Sec. VI A limits the interpretation of Table II. Appendices A–D address parametric-overfitting, 3D/2D, and benchmark-scope critiques raised in peer review.

ACKNOWLEDGMENTS

The author expresses gratitude to the Planck Collaboration, the SLACS team, the CASTLES survey, and the LIGO-Virgo-KAGRA Scientific Collaboration for maintaining publicly accessible observational catalogs and data repositories. Numerical reproducibility uses the `cromatic2` electromagnetic channel package [38] and the `GRAVEDAD3` tensor / host validation tree [39].

Appendix A: Cubic truncation and the ratio ζ/χ

a. Mechanical–entropic dictionary. At the P_0 branch of Eq. (16), $V''(P_0) = 2\lambda_P P_0^2$ and $V'''(P_0) = 6\lambda_P P_0$. The GRAVEDAD3 thermodynamic reading identifies $\chi = T_{\rm GH}^{-1} V''(P_0)$ on the passive shell. The cubic entropy coefficient is *not* tuned to Planck; it is fixed by the one-sided activation threshold $\phi_0 < P_0$ of the `VACIO` companion: nucleation/cooperativity turns on only for $P \gtrsim \phi_0$, breaking the reflection symmetry $(P - P_0) \to -(P - P_0)$ of the quadratic well. The leading odd correction is therefore $\mathcal{O}((P - P_0)^3)$; even $(P - P_0)^4$ terms are subleading for IR shells of thickness $\Delta_P \leq \phi_0$.

b. Mechanical upper bound versus activation reduction. At the P_0 branch of Eq. (16), $V''(P_0) = 2\lambda_P P_0^2$, $V'''(P_0) = 6\lambda_P P_0$, and $\chi = T_{\rm GH}^{-1} V''(P_0)$. A naive Gibbs shell balance $G(P_0 + \Delta_P) - G(P_0) = 0$ with $\Delta_P = \phi_0$

would set $\zeta = V'''(P_0)/T_{\rm GH}$ and hence $\zeta/\chi = 3/P_0 \simeq 15.6$, which *overshoots* the attractor ($x_\star \gg 1$). The `VACIO` one-sided activation threshold $\phi_0 < P_0$ breaks $(P-P_0) \to -(P-P_0)$ symmetry and suppresses the odd entropy slope by the fractional gap $\eta = (P_0-\phi_0)/P_0$ over the shell thickness ϕ_0:

$$\frac{\zeta}{\chi} = \frac{\eta}{3\phi_0}, \qquad \chi = \frac{2\lambda_P P_0^2}{T_{\rm GH}}, \tag{A1}$$

i.e. Eq. (39) with $\Delta_P = \phi_0$. Relative to the mechanical estimate $3/P_0$, this is smaller by $\sim \eta P_0/9 \simeq 1\%$; the Planck-near closure therefore uses a *strongly activation-suppressed* cubic slope, not the raw $V'''(P_0)$ shell balance alone. The factor 1/3 in Eq. (A1) is the first-order projection of $\chi_{\rm eff} = \chi - \zeta\Delta_P$ onto $\mathcal{L}_{\rm loc}$, yielding Eq. (40).

c. Why stop at cubic order? The next term $-(\xi/24)(P-P_0)^4$ contributes on the shell at relative size $\sim |\xi|\Delta_P/(4|\zeta|)$. With the activation-reduced ζ of Eq. (A1) and the mechanical estimate $\xi \sim V''''(P_0)/T_{\rm GH} = 6\lambda_P/T_{\rm GH}$, the audit script gives $|\xi|\Delta_P/(4|\zeta|) \sim 1.3$ on the shell. Quartic corrections are therefore *comparable* to cubic ones at $\Delta_P = \phi_0$, not negligible at the permille level. The +0.17% Planck agreement must therefore be read as a *numerically narrow post-diction* within a truncated EFT, not as a unique microscopic prediction. Stopping at cubic order is a *working truncation* justified by the leading broken-symmetry odd term, not by a small-parameter expansion in $\Delta_P/\ P_0$.

d. Overfitting audit (script `camino12_Svac_cubic.py`). Inputs frozen *before* the χ closure: $P_0 = 0.1921$, $\phi_0 = 0.10$, $R_{\rm crit} = (P_0/\phi_0)^2$, and H_0. No parameter is adjusted to minimize $|x_\star - \Omega_\Lambda|$. Varying ζ/χ by $\pm 20\%$ shifts $x_\star$ by $\mp 0.4\%$ (Table IV), so the closure is testable but the Planck agreement is *numerically narrow.*

TABLE IV. Sensitivity of $x_\star$ to ζ/χ (overfitting audit).

ζ/χ factor	$x_\star$	Δ vs. Planck
0.80× nominal	0.685	−0.38%
nominal ($\eta/3\phi_0$)	0.689	+0.17%
1.20× nominal	0.693	+0.72%

Appendix B: Three-dimensional bulk hysteresis as a null test

The `VACIO` PDE is a phenomenological continuum model. When run as a volumetric 48^3 hysteresis cycle without horizon projection, it returns $f_{\rm hyst}^{\rm 3D} \simeq 0.92$, i.e. +34% above the analytic attractor $x_\star \simeq 0.67$–0.69 and +23% above Planck Ω_Λ. This is *not* treated as a failed prediction to be tuned away; it is a null test of the hypothesis "Λ is stored as bulk volume memory." The covariant closure Eqs. (33)–(37) couple the organized fraction to $\sigma_{\rm eff}A_H$ (area), not to $\rho_{\rm crit}c^2V_H$ with an unprojected bulk fraction. Dimensional analysis therefore *requires* a holographic/boundary readout for Ω_Λ-scale hysteresis [14, 16]; the 3D overshoot is the expected failure mode of the alternative hypothesis. The aligned 2D run ($f_{\rm hyst}^{\rm 2D} \simeq 0.657$–$0.672$) is the compatible channel, not an ad hoc discard of inconvenient code.

Appendix C: Deep-field MOND limit of Eq. (17)

Test 2 does *not* claim that Eq. (17) has been integrated over SPARC galaxies. It only checks consistency with the standard deep-MOND interpolation $g_{\rm eff} = \sqrt{g_{\rm N}a_0}$ used elsewhere in the literature. A derivation path *within* the EFT is nevertheless identifiable. In the static weak-field limit with $\alpha_A(P_0) = 0$, Eq. (25) links a_0 to the exterior gradient $P_{\rm vac} = P_0 - C/r$ via $K_gC = r_*^2 a_0$. In the chameleon/thin-shell regime (Sec. III A), $|\delta P/P_0| \ll 1$ in the Solar System, but at galactic radii where $g_{\rm N} \lesssim a_0$ the nonlinear drive $-4\pi G\lambda_v T_{\rm eff}$ in Eq. (17) competes with the gradient term. Matching the transition radius where $K_g|\nabla P_{\rm vac}| = g_{\rm N}$ to the saturation scale $K_g|\nabla P_{\rm vac}| = a_0$ yields, in the deep limit $g_{\rm N} \ll a_0$,

$$g_{\rm eff} \sim \sqrt{g_{\rm N}a_0}, \tag{C1}$$

the QUMOND/deep-MOND scaling, *provided* $C \propto \sqrt{M}$ in the low-acceleration branch (standard chameleon screening). This is a sketch, not the numerical SPARC proof requested for publication-grade Test 2; it removes the charge of an *unmotivated* external formula while keeping the benchmark scope honest. Script `test2_sparc_field.py` integrates the spherical AQUAL limit $\nabla\cdot(\mu(g/a_0)\mathbf{g}) = 4\pi G\rho$ with $\mu(x) = x/\sqrt{1+x^2}$ (no external $\sqrt{g_N a_0}$ input) [47–49]. For $M = 10^{11}M_\odot$, $R_d = 3\,$kpc, $v(50\,{\rm kpc}) \simeq 201\,$km/s ($\sim 4\%$ mean offset from the MOND interpolation benchmark; suite band 150–250 km/s passed).

Appendix D: Core saturation from Eq. (16)

Test 4 does not solve the full coupled system $G_{\mu\nu} = T_{\mu\nu}^{\rm vac}(P_{\rm vac}) + T_{\mu\nu}^{\rm matter}$ in spherical symmetry. It implements a bounded pressure proxy tied to the saturating branch of the *same* potential Eq. (16). When local curvature/ density raises the effective potential energy density toward $\rho_{\rm loc}c^2$, the quartic well forces a finite excursion $\delta P_{\rm sat}$ via $V(P_0 + \delta P_{\rm sat}) \sim \rho_{\rm loc}c^2$, i.e.

$$\delta P_{\rm sat} \sim \sqrt{\frac{2\rho_{\rm loc}c^2}{\lambda_P P_0^2}} \quad \text{(small-}\delta P\text{ branch).} \tag{D1}$$

Equation (41) is the phenomenological inversion of this saturating branch in the $r \to 0$ gauge (finite $P_{\rm vac}$ instead of divergent gradient). Script `test4_spherical_core.py` integrates the Hayward-compatible system with $\lambda_P = 3/(2\pi\ell_{\rm sat}^2 P_0^4)$ derived from the quartic core (see reference/attack_gr_bh_derived.py and Ref. [50]): $K(0)$ remains finite and $\ell_{\rm sat}$ is *not* postulated from Eq. (41). A publication-grade upgrade—still open—is a full coupled Einstein–$P_{\rm vac}$ integration replacing the Hayward ansatz for $f(r)$.

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